

Haseeb Ul Hassan

AI / Machine Learning Engineer

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SUMMARY

Machine Learning Engineer at VisionX, building and deploying AI products for enterprise clients. Hands-on experience across Computer Vision, NLP, and Generative AI, with a track record of fine-tuning models, automating data pipelines, and delivering solutions such as ForecastIQ that saved \$100K+ annually and reduced operational overhead.

WORK EXPERIENCE

VisionX

Islamabad, Pakistan

Junior Machine Learning Engineer

Nov 2024 – Present

- Develop custom AI and ML solutions for enterprise clients, including data augmentation and model fine-tuning to improve model performance and delivery speed.
- Built ForecastIQ, an end-to-end demand-forecasting and planning pipeline that centralized inputs, automated daily processing, and eliminated manual spreadsheet work, saving \$100K+/year in labor and reducing excess inventory.
- Lead ongoing ML initiatives across PromoIQ, automation, and deployment workstreams to improve operational efficiency and speed up delivery of client-facing solutions.
- Led and mentored a team of ML interns, ran daily standups, assigned work, and conducted weekly demos to improve execution speed and build team capability.
- Delivered custom AI solutions for Kofile, automating image-based cataloging and data extraction across multiple categories to reduce manual entry, improve throughput, and increase data accuracy.
- Manage and update APIs for ongoing AI services, ensuring seamless integration and functionality.
- Implemented generative AI workflows for client and internal use cases, enabling automation and expanding product capabilities across internal tools and client deliverables.

Machine Learning Intern

Aug 2024 – Oct 2024

- Trained PaddleOCR Detection model by converting custom dataset labels, enhancing text detection performance.
- Fine-tuned LayoutLMv3 for sequence classification, tuning batch size and learning rate to improve model accuracy on document understanding tasks.
- Trained and compared detection and segmentation models, converted segmentation labels to detection format, and applied augmentation techniques to improve accuracy and generalization.
- Developed and tested Android-based mobile applications integrating computer vision models for detection and segmentation workflows.

EDUCATION

Institute of Space Technology (IST)

Islamabad, Pakistan

B.S. in Computer Science CGPA: 3.77 / 4.0 (Presidential Gold Medalist)

Sep 2020 – July 2024

SELECTED CERTIFICATIONS FROM COURSERA

IBM – Building Deep Learning Models with TensorFlow

IBM – Introduction to Deep Learning & Neural Networks with Keras

IBM – Python for Data Science, AI & Development

Multiple Partners – Supervised Machine Learning: Regression and Classification

DeepLearning.AI – AI For Everyone

SKILLS

Languages & Frameworks: Python, C++, SQL, PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Ultralytics, Transformers, LangChain, YOLO, Roboflow, FastAPI, Docker, AWS, Git, Linux, Weights & Biases, n8n.

SELECTED PROJECTS

Automated Book Spine Detection and Information Extraction System

- Developed and implemented a pipeline combining Computer Vision (CV) and Natural Language Processing (NLP) to extract and classify information from book spines in images.
- Utilized OCR to extract text from images and trained a custom YOLO OBB model achieving high accuracy in detecting rotated book spines.
- Integrated a custom Named Entity Recognition (NER) model to classify extracted text into predefined classes, displaying results in a structured format, and LLM call for customization and accuracy.

ForecastIQ

- Designed, built, and deployed ForecastIQ, an end-to-end ML pipeline (feature engineering, model training, and automated Python planning) that replaced ad-hoc ordering, saving ~\$100K+ annually in labor and reducing excess inventory, realizing multi-million dollar savings and freeing capital for reinvestment.

NORO AI

- AI Assistant with full-featured Institutional-Grade Trading knowledge, integrated with real-time market data APIs (Binance, Alpha Vantage, GoldAPI) and configurable web search providers for live market research.
- Developed and integrated Telegram bot with webhook handling, session mapping, and multimedia (image) support for seamless mobile access.

Brain Tumor Classification for Healthcare Services

- Proposed, Developed and fine-tuned custom Convolutional Neural Network (CNN) architecture.
- Preprocessed and augmented the dataset, tuning model hyperparameters for optimal performance.
- Comparison of proposed CNN with other Deep Learning models and Machine Learning models.